

Vis-OntoQ: An Ontology-Guided Framework to Support Non-Expert Analysts Formulating Data Questions during Visual Exploration

Dieinison Jack Freire Braga
Department of Informatics
Pontifical Catholic University of Rio de Janeiro
Rio de Janeiro, RJ, Brazil
dbraga@inf.puc-rio.br

Abstract

Formulating relevant data questions is a critical but challenging step in visual data exploration, especially for non-expert analysts. While modern visualization tools can support exploration, they often lack domain-specific knowledge, leading to generic or untrustworthy outputs. Since LLMs alone cannot fully represent domain semantics, ontological grounding provides the referential mapping needed to anchor generated questions in domain-relevant entities and relations. This thesis proposes a framework, named Vis-OntoQ, that integrates domain ontologies into a Retrieval-Augmented Generation pipeline to guide data question formulation during visual data exploration. Vis-OntoQ will be integrated into a mixed-initiative visualization system to assist and collaborate with analysts in their visual data exploration. We combine a systematic literature review, ontology construction, integration into a visualization system, and user evaluation to assess how ontology-grounded question generation influences analysts' visual exploration workflows.

CCS Concepts

• **Human-centered computing** → **Empirical studies in visualization**; **Visualization systems and tools**.

Keywords

Human-centered AI, Question formulation, Visual exploration, Novice analysts, Ontology-guided systems

ACM Reference Format:

Dieinison Jack Freire Braga. 2026. Vis-OntoQ: An Ontology-Guided Framework to Support Non-Expert Analysts Formulating Data Questions during Visual Exploration. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3799218>

1 Introduction

Tukey defined exploratory data analysis as the process of learning about and making sense of a new data set [48]. In particular, visual data exploration enables analysts¹ to uncover insights [19, 55], inspect visualized data [7], and iteratively refine their questions and hypotheses [2, 47]. Although most modern data analysis tools emphasize a flexible approach to open-ended exploratory data analysis, they often leave the exploration process unguided, which can be challenging for non-expert analysts² [3, 25, 36, 44, 50, 52, 53].

In the early stages of the visual exploration, analysts formulate questions along a spectrum, ranging from (1) having no data questions at all, through (2) ill-defined data questions, to (3) clearly defined data questions [2]. While one of the main strengths of visualization lies not in answering questions, but in supporting analysts to formulate questions [49], non-expert analysts often fall within the two initial stages of this spectrum, *i.e.*, they might not know what data questions to ask [12, 31] or they may pose ill-defined questions [5, 29].

Recently, a growing number of *Human-centered AI* [40] tools have been proposed, aiming to amplify and augment human performance using AI models. In this direction, visualization researchers are investigating the combination of Generative AI (GenAI) capabilities with human expertise to support visual analysis processes, such as pattern discovery [54], hypothesis generation [16], and others [9, 11]. However, there is still a lack of understanding and support for the data questions formulation process, which is critical to guide effective visualization design [24]. Supporting data question generation is challenging because this phase of the analysis process is an inherently human activity [51], and analysts use a wide variety of approaches to visually explore the data [18].

Moreover, many current GenAI-based approaches still operate largely on generic language models, without incorporating the domain knowledge underlying the data [1]. Recent work argues that LLMs simulate rather than truly represent domain semantics, *i.e.*, whatever knowledge they acquire is buried in statistical patterns that humans cannot meaningfully interpret, making explainability difficult to attain [45]. This disconnection can lead to a weak grounding, thus increasing the risk of hallucinations and undermining analysts' trust [20]. The grounding of LLM outputs in formal



This work is licensed under a Creative Commons Attribution 4.0 International License. *CHI EA '26, Barcelona, Spain*

© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2281-3/26/04
<https://doi.org/10.1145/3772363.3799218>

¹We refer *analyst* to those who use visualization tools in any capacity, not limited to a specific person or role.

²We use the term *non-expert* for analysts who lack visualization-specific knowledge in both theoretical concepts and evaluation methods.

ontological structures has been identified as a promising direction to address this limitation, as ontologies provide the referential mapping needed to anchor model output in domain-relevant entities and relations [14, 45]. In this direction, we can frame domain ontologies [13, 42, 43] as knowledge bases in a RAG pipeline to potentially support data question formulation. In this thesis, we aim to investigate the use of domain ontology as a semantic backbone in a RAG pipeline to generate data questions grounded in the relevant entities and relations [10, 27, 32, 34, 41].

In this context, we propose a framework, named Vis-OntoQ, to support non-experts in formulating and refining data questions during visual data exploration. Vis-OntoQ will consist of: (1) a systematic synthesis of how the visualization research community defines, supports, and evaluates data questions; (2) an ontology-guided question formulation model that aligns analyst-provided prompts to data questions; and (3) empirical evidence on how domain ontologies can support the visual exploration process.

2 Research Claim and Proposal

Our plan with this thesis is to leverage the complementary strengths of human analytical reasoning and Gen-AI capabilities to support non-expert analysts in visual data exploration. For that, we defined our main research question: *How can ontologies be used to support non-expert analysts in formulating questions in visual data exploration?*

- **RQ1:** How does the visualization literature define, support, and evaluate data questions?
- **RQ2:** How can ontologies support data question formulation?
- **RQ3:** How does the proposed ontology approach transform visual data exploration?

3 Methods

In this section, we outline the methods to address each of our research questions. Figure 1 presents an overview of the steps proposed to carry out this thesis. We detail each step below.

Specifically for RQ1, we conducted an initial exploratory study (Step 1) to understand how non-expert analysts perceived the usefulness of a question-oriented visual analysis approach [5, 26]. In addition, we are conducting a systematic literature review (Step 2) [21, 30] to synthesize: definitions of data questions, proposed approaches to generate and refine data questions, and evaluation practices (study designs, metrics, and others). By the end of these two steps, we aim to provide a structured view of what a “data question” is and how non-experts perceive it in practice.

For RQ2, we will experiment to use an RAG model (Step 3) in three phases: (i) to generate a domain ontology guided by the consolidated Sowa’s model, (ii) to serve this domain ontology as a semantic backbone in a RAG pipeline, and (iii) embed this RAG model in a mixed-initiative visualization system to generate relevant data questions [4, 17]. Concretely, the RAG model will receive the analysts’ prompt-based analytical questions. The RAG model will then reason within the domain ontology to align the analyst’s prompt with semantically richer data questions. Moreover, this

approach could foster explainability through provenance and traceability, as the rationale behind the data questions generated are accessible through the ontology [42].

To address RQ3, we will use the ontology-grounded RAG model embedded in the data visualization system to evaluate the proposed ontology-driven approach with non-expert analysts through a mixed-methods user study (Step 4) – think-aloud protocols, interaction logs, and post-study measures [8, 23, 28, 38]. This phase may be adjusted based on findings from RQ1 and RQ2 to effectively assess how users iteratively generate and refine data questions. For instance, we can frame this phase based on Schön’s notion of reflective conversation, in which the analyst expresses an analytical question, the RAG model responds, then the analyst reflects and replies. The ontology-grounded visualization system “talks back,” prompting further reflection and action. The goal is to turn ill-defined prompt-based data questions into specific, relevant ones through a side-by-side collaboration [11, 37].

4 Results and Contributions to Date

4.1 Step 1: Question-Oriented Visual Exploration Approach

Addressing RQ1, we conducted an initial exploratory investigation, examining a question-guided visualization recommendation approach to assist non-expert analysts in exploratory analysis. Our findings indicate that non-expert analysts perceived system-generated questions as useful in leading to insights, generating new hypotheses, and supporting the data exploration process overall [5].

While these results underscore the feasibility of question-based support, they also revealed limitations in template-based generation [33]. These limitations thus motivate our ontology-grounded approach presented in RQ2.

4.2 Step 2: Systematic Literature Review of Data Questions

Also, to address RQ1, we are currently conducting a systematic literature review of how the visualization community addresses data questions. We are in the middle of the full-text data extraction phase. Preliminary results suggest a range of complementary approaches summarized below:

- Suh et al. present a grammar for expressing hypotheses in visual data analysis that operationalizes users’ data analysis questions as a hypothesis space [46];
- Query-formulation assistance for analytical natural language interaction, e.g., autocompletion designed to reduce the cognitive load of formulating analytical queries [39];
- Natural Language interfaces that decompose questions to support exploratory visual analysis [15];
- Automated exploratory data analysis workflows that are explicitly question-guided [24].

This review will consolidate definitions of “data questions” and highlight open challenges that motivate our ontology-grounded RAG approach for generating data questions.

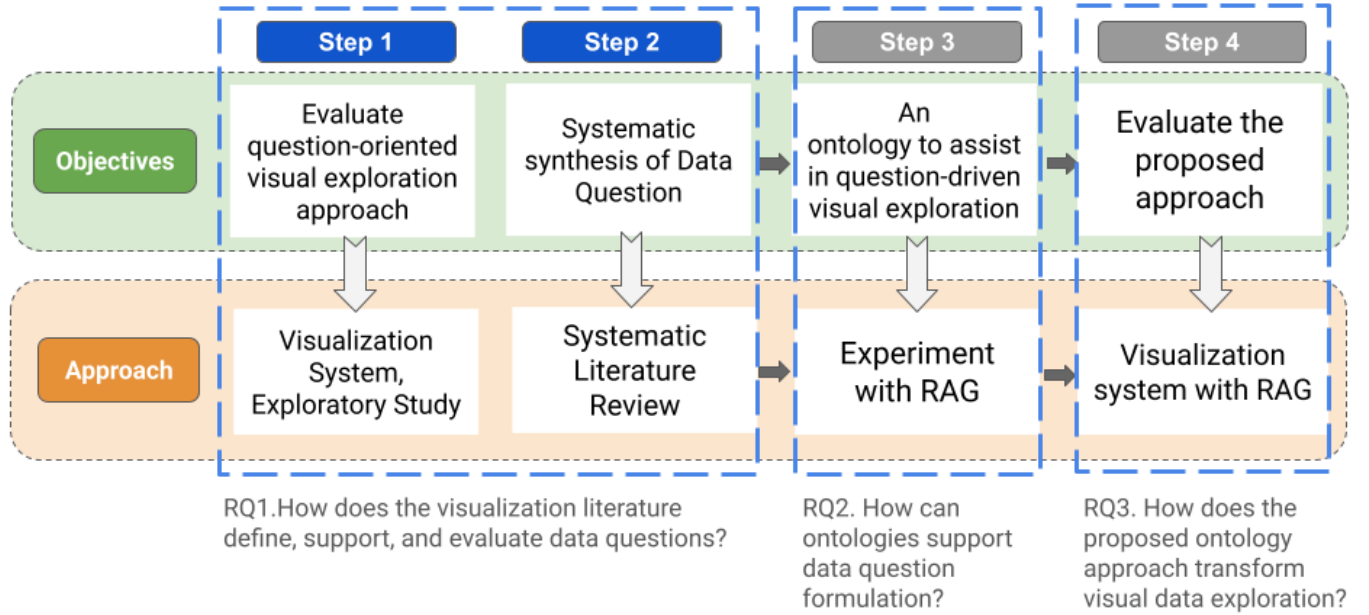


Figure 1: Overview of research to date and expected next steps to address research questions.

5 Next Steps

5.1 Step 3: Experiment with Retrieval-Augmented Generation

Our next steps involve a two-stage model-building phase. In the first stage, we plan to experiment with using a RAG model to construct a domain ontology from a dataset of scientific manuscript discussions, where the analytical task involves comparing rhetorical structures across texts, such as how claims, evidence, and research gaps are distributed and related across forum submissions. To consolidate the candidate concepts and relations extracted, we intend to ground the ontology schema in Guizzardi et al.’s Unified Foundational Ontology (UFO), which provides formal distinctions between types, instances, and relations. We are motivated by recent arguments that LLMs alone cannot fully manage domain semantics, and that formal ontological grounding may help reduce ambiguity in retrieval and generation [45]. By representing rhetorical entities, such as claims, evidence, and contributions, as formally grounded universals with well-defined semantic relations, we hope to address this limitation in a principled way.

In the second stage, this ontology will serve as a semantic backbone for a second RAG model embedded within Arcano [4], a mixed-initiative visualization system that allows analysts to author visualizations and browse a gallery of system-generated data questions alongside recommended visualizations. The RAG model will reason on the domain ontology to retrieve and generate domain-relevant data questions, serving as the experimental scenario for the user evaluation in Step 4.

5.2 Step 4: Evaluation of the Proposed Ontology-grounded RAG Model

To evaluate the integrated ontology-grounded RAG model, we will follow a mixed-methods approach [8], combining (i) perceived value measures, such as questionnaires and tool logs, with (ii) qualitative insights from semi-structured interviews [23]. The overall goal is to understand whether and how an ontology approach can transform a typical visual data exploration process. We plan to recruit participants as domain experts, *i.e.*, analysts with deep domain knowledge but without specialized visualization training, who struggle to independently formulate data questions and assess which visualization is most appropriate for those questions.

Quantitative measures: We will instrument the data visualization system to capture interaction logs, in combination with questionnaires measuring perceived value:

- Number of formulated questions during the session [22].
- Number of findings extracted from sessions (*e.g.*, through think-aloud notes) [35].

Qualitative measures: To explain *why* the model supports (or fails to support), we will conduct post-session semi-structured interviews and analyze them with thematic analysis [6].

6 Research Status and Long-Term Goals

I am a second-year PhD student in the Department of Informatics at Pontifical Catholic University of Rio de Janeiro (PUC-Rio), Brazil. My advisor is Professor Simone D. J. Barbosa. I have completed all required coursework and have passed our program’s qualification exam. As part of the standard four-year PhD program at PUC-Rio, my expected completion date is December 2027. My long-term goal is to conduct research in human-centered AI (HCAI) and data visualization, exploring AI’s potential for fostering visual analysis

and analytics. Upon graduation, I plan to seek research positions in academia and industry. I look forward to completing my thesis with feedback from the Student Mentoring Program, which I have not yet had the opportunity to participate in.

Acknowledgments

This study was financed in part by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior, Brazil (CAPES), Finance code 001.

References

- [1] Rahul C. Basole and Timothy Major. 2024. Generative AI for Visualization: Opportunities and Challenges. *IEEE Computer Graphics and Applications* 44, 2 (March 2024), 55–64. doi:10.1109/MCG.2024.3362168
- [2] Leilani Battle and Jeffrey Heer. 2019. Characterizing Exploratory Visual Analysis: A Literature Review and Evaluation of Analytic Provenance in Tableau. *Computer Graphics Forum* 38, 3 (June 2019), 145–159. doi:10.1111/cgf.13678
- [3] M. Bostock, V. Ogievetsky, and J. Heer. 2011. D³ Data-Driven Documents. *IEEE Transactions on Visualization and Computer Graphics* 17, 12 (Dec. 2011), 2301–2309. doi:10.1109/TVCG.2011.185
- [4] Deinison J. F. Braga, Raul A. Lima, João D. L. dos Santos, George M. de Oliveira, Marisa C. da Silva, and Simone D. J. Barbosa. 2025. Arcano: Generating Data Questions to Support Non-Expert Analysts in Hypothesis Formulation. In *Human-Computer Interaction – INTERACT 2025: 20th IFIP TC 13 International Conference, Belo Horizonte, Brazil, September 8–12, 2025, Proceedings, Part IV*. Springer-Verlag, Berlin, Heidelberg, 100–105. doi:10.1007/978-3-032-05008-3_20
- [5] Deinison Jack Freire Braga, Marisa Carmo da Silva, Raul de Araújo Lima, Gabriel Diniz Junqueira Barbosa, and Simone Diniz Junqueira Barbosa. 2024. VisStoryMaker: supporting non-expert analysts in visually exploring datasets and communicating insights with visual annotations and data stories. In *Proceedings of the XXIII Brazilian Symposium on Human Factors in Computing Systems (IHC '24)*. Association for Computing Machinery, New York, NY, USA, 1–15. doi:10.1145/3702038.3702113
- [6] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology* 3, 2 (Jan. 2006), 77–101. doi:10.1191/1478088706qp063oa
- [7] William S. Cleveland. 1985. *The elements of graphing data*. Wadsworth Publ. Co., Monterey, CA. <https://dl.acm.org/doi/abs/10.5555/4084>
- [8] John W. Creswell and J. David Creswell. 2017. *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches*. SAGE Publications, Thousand Oaks, CA.
- [9] Vaishali Dhanoo, Anton Wolter, Gabriela Molina León, Hans-Jörg Schulz, and Niklas Elmqvist. 2025. Agentic Visualization: Extracting Agent-Based Design Patterns From Visualization Systems. *IEEE Computer Graphics and Applications* 45, 6 (Nov. 2025), 89–100. doi:10.1109/MCG.2025.3607741
- [10] D.J. Duke, K.W. Brodli, and D.A. Duce. 2004. Building an Ontology of Visualization. In *IEEE Visualization 2004*. IEEE, Austin, TX, USA, 7p–7p. doi:10.1109/VISUAL.2004.10
- [11] Niklas Elmqvist and Clemens Nylandstedt Klokmoose. 2025. Automating the Path: An R&D Agenda for Human-Centered AI and Visualization. *IEEE Computer Graphics and Applications* 45, 03 (May 2025), 73–81. doi:10.1109/MCG.2025.3559374
- [12] Lars Grammel, Melanie Tory, and Margaret-Anne Storey. 2010. How Information Visualization Novices Construct Visualizations. *IEEE Transactions on Visualization and Computer Graphics* 16, 6 (Nov. 2010), 943–952. doi:10.1109/TVCG.2010.164
- [13] Thomas R. Gruber. 1993. A translation approach to portable ontology specifications. *Knowledge Acquisition* 5, 2 (June 1993), 199–220. doi:10.1006/knac.1993.1008
- [14] Giancarlo Guizzardi, Alessandro Botti Benevides, Claudenir M Fonseca, Daniele Porello, João Paulo A Almeida, and Tiago Prince Sales. 2022. UFO: Unified foundational ontology. *Applied ontology* 17, 1 (2022), 167–210.
- [15] Yi Guo, Danqing Shi, Mingjuan Guo, Yanqiu Wu, Nan Cao, and Qing Chen. 2024. Talk2Data: A Natural Language Interface for Exploratory Visual Analysis via Question Decomposition. *ACM Trans. Interact. Intell. Syst.* 14, 2 (2024), 8:1–8:24. doi:10.1145/3643894
- [16] Md Naimul Hoque, Sungbok Shin, and Niklas Elmqvist. 2024. Visualization for Human-Centered AI Tools. doi:10.48550/arXiv.2404.02147
- [17] Eric Horvitz. 1999. Principles of mixed-initiative user interfaces. In *Proceedings of the SIGCHI conference on Human factors in computing systems the CHI is the limit - CHI '99*. ACM Press, Pittsburgh, Pennsylvania, United States, 159–166. doi:10.1145/302979.303030
- [18] Petra Isenberg, Anthony Tang, and Sheelagh Carpendale. 2008. An exploratory study of visual information analysis. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '08)*. Association for Computing Machinery, New York, NY, USA, 1217–1226. doi:10.1145/1357054.1357245
- [19] T.j. Jankun-Kelly, Kwan-liu Ma, and Michael Gertz. 2007. A Model and Framework for Visualization Exploration. *IEEE Transactions on Visualization and Computer Graphics* 13, 2 (March 2007), 357–369. doi:10.1109/TVCG.2007.28
- [20] Haoran Jiang, Shaohan Shi, Yunjie Yao, Chang Jiang, and Quan Li. 2025. HypoChainer: a Collaborative System Combining LLMs and Knowledge Graphs for Hypothesis-Driven Scientific Discovery. *IEEE Transactions on Visualization and Computer Graphics* 32, 1 (2025), 1–11. doi:10.1109/TVCG.2025.3633887
- [21] Barbara Kitchenham and Stuart Charters. 2007. *Guidelines for Performing Systematic Literature Reviews in Software Engineering*. Technical Report EBSE-2007-01. Keele University, Keele, UK.
- [22] Heidi Lam, Enrico Bertini, Petra Isenberg, Catherine Plaisant, and Sheelagh Carpendale. 2012. Empirical Studies in Information Visualization: Seven Scenarios. *IEEE Transactions on Visualization and Computer Graphics* 18, 9 (Sept. 2012), 1520–1536. doi:10.1109/TVCG.2011.279
- [23] Jonathan Lazar, Jinjuan Heidi Feng, and Harry Hochheiser. 2017. *Research Methods in Human-Computer Interaction*. Morgan Kaufmann, San Francisco, CA.
- [24] Elsie Lee-Robbins, Shiqing He, and Eytan Adar. 2022. Learning Objectives, Insights, and Assessments: How Specification Formats Impact Design. *IEEE Transactions on Visualization and Computer Graphics* 28, 1 (Jan. 2022), 676–685. doi:10.1109/TVCG.2021.3114811
- [25] Deqing Li, Honghui Mei, Yi Shen, Shuang Su, Wenli Zhang, Junting Wang, Ming Zu, and Wei Chen. 2018. ECharts: A declarative framework for rapid construction of web-based visualization. *Visual Informatics* 2, 2 (June 2018), 136–146. doi:10.1016/j.visinf.2018.04.011
- [26] Raul de Araújo Lima and Simone Diniz Junqueira Barbosa. 2020. A Question-Oriented Visualization Recommendation Approach for Data Exploration. In *Proceedings of the 2020 International Conference on Advanced Visual Interfaces (AVI '20)*. Association for Computing Machinery, New York, NY, USA, 1–5. doi:10.1145/3399715.3399849
- [27] Suzana Loshkovska and Panče Panov. 2025. Foundations for a Generic Ontology for Visualization: A Comprehensive Survey. *Information* 16, 10 (Oct. 2025), 915. doi:10.3390/info16100915
- [28] Tamara Munzner. 2009. A Nested Model for Visualization Design and Validation. *IEEE Transactions on Visualization and Computer Graphics* 15, 6 (Nov. 2009), 921–928. doi:10.1109/TVCG.2009.111
- [29] Tamara Munzner. 2014. *Visualization Analysis and Design* (1 ed.). A K Peters/CRC Press, Boca Raton, FL. doi:10.1201/b17511
- [30] Kai Petersen, Sairam Vakkalanka, and Ludwik Kuzniarz. 2015. Guidelines for conducting systematic mapping studies in software engineering: An update. *Information and Software Technology* 64 (Aug. 2015), 1–18. doi:10.1016/j.infsof.2015.03.007
- [31] Peter Pirollo and Stuart Card. 2005. The sensemaking process and leverage points for analyst technology as identified through cognitive task analysis. In *Proceedings of international conference on intelligence analysis*, Vol. 5. Office of the Director of National Intelligence, McLean, VA, USA, 2–4.
- [32] Jan Polowinski and Martin Voigt. 2013. VISO: a shared, formal knowledge base as a foundation for semi-automatic infovis systems. In *CHI '13 Extended Abstracts on Human Factors in Computing Systems (CHI EA '13)*. Association for Computing Machinery, New York, NY, USA, 1791–1796. doi:10.1145/2468356.2468677
- [33] Ariane Moraes Bueno Rodrigues, Gabriel Diniz Junqueira Barbosa, Hélio Côrtes Vieira Lopes, and Simone Diniz Junqueira Barbosa. 2021. What questions reveal about novices' attempts to make sense of data visualizations: Patterns and misconceptions. *Computers & Graphics* 94 (Feb. 2021), 32–42. doi:10.1016/j.cag.2020.09.015
- [34] Dominik Sacha, Matthias Kraus, Daniel A. Keim, and Min Chen. 2019. VIS4ML: An Ontology for Visual Analytics Assisted Machine Learning. *IEEE Transactions on Visualization and Computer Graphics* 25, 1 (Jan. 2019), 385–395. doi:10.1109/TVCG.2018.2864838
- [35] Ali Sarvghad, Melanie Tory, and Narges Mahyar. 2017. Visualizing Dimension Coverage to Support Exploratory Analysis. *IEEE Transactions on Visualization and Computer Graphics* 23, 1 (2017), 21–30. doi:10.1109/TVCG.2016.2598466
- [36] Arvind Satyanarayan, Dominik Moritz, Kanit Wongsuphasawat, and Jeffrey Heer. 2017. Vega-Lite: A Grammar of Interactive Graphics. *IEEE Transactions on Visualization and Computer Graphics* 23, 1 (Jan. 2017), 341–350. doi:10.1109/TVCG.2016.2599030
- [37] Donald A. Schön. 1983. *The reflective practitioner: how professionals think in action*. Basic Books, New York. OCLC: 1484614411.
- [38] Michael Sedlmair, Miriah Meyer, and Tamara Munzner. 2012. Design Study Methodology: Reflections from the Trenches and the Stacks. *IEEE Transactions on Visualization and Computer Graphics* 18, 12 (Dec. 2012), 2431–2440. doi:10.1109/TVCG.2012.213
- [39] Vidya Setlur, Enamul Hoque, Dae Hyun Kim, and Angel X. Chang. 2020. Sneak Pique: Exploring Autocompletion as a Data Discovery Scaffold for Supporting Visual Analysis. In *Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology (UIST '20)*. Association for Computing Machinery, New York, NY, USA, 966–978. doi:10.1145/3379337.3415813

- [40] Ben Shneiderman. 2022. *Human-centered AI*. Oxford University Press, Oxford, UK.
- [41] Gao Shu, Nick J. Avis, and Omer F. Rana. 2008. Bringing semantics to visualization services. *Advances in Engineering Software* 39, 6 (June 2008), 514–520. doi:10.1016/j.advengsoft.2007.03.016
- [42] John F. Sowa. 1999. *Knowledge representation: logical, philosophical and computational foundations*. Brooks/Cole Publishing Co., Pacific Grove, CA. <https://dl.acm.org/doi/abs/10.5555/318183>
- [43] Steffen Staab and Rudi Studer. 2010. *Handbook on Ontologies*. Springer Science & Business Media, Berlin, Germany.
- [44] Chris Stolte, Diane Tang, and Pat Hanrahan. 2008. Polaris: a system for query, analysis, and visualization of multidimensional databases. *Commun. ACM* 51, 11 (Nov. 2008), 75–84. doi:10.1145/1400214.1400234
- [45] Veda C Storey, Oscar Pastor, Giancarlo Guizzardi, Stephen W Liddle, Wolfgang Maaß, Jeffrey Parsons, Jolita Ralyté, and Maribel Yasmina Santos. 2025. Large language models for conceptual modeling: Assessment and application potential. *Data & Knowledge Engineering* 160 (2025), 102480.
- [46] Ashley Suh, Ab Mosca, Eugene Wu, and Remco Chang. 2023. A Grammar of Hypotheses for Visualization, Data, and Analysis. doi:10.48550/arXiv.2204.14267
- [47] James J. Thomas. 2005. *Illuminating the path: the Research and Development Agenda for Visual Analytics*. IEEE Computer Society, Los Alamitos, CA.
- [48] John W. Tukey. 1977. *Exploratory Data Analysis*. Addison-Wesley, Reading, MA.
- [49] John W. Tukey. 1980. We Need Both Exploratory and Confirmatory. *The American Statistician* 34, 1 (Feb. 1980), 23–25. doi:10.1080/00031305.1980.10482706
- [50] Jacob VanderPlas, Brian Granger, Jeffrey Heer, Dominik Moritz, Kanit Wongsuphasawat, Arvind Satyanarayan, Eitan Lees, Ilia Timofeev, Ben Welsh, and Scott Sievert. 2018. Altair: interactive statistical visualizations for Python. *Journal of open source software* 3, 32 (2018), 1057. <http://www.theoj.org/joss-papers/joss.01057/10.21105.joss.01057.pdf>
- [51] Colin Ware. 2010. *Visual thinking for design*. Elsevier, Burlington, MA.
- [52] Hadley Wickham. 2011. ggplot2. *WIREs Computational Statistics* 3, 2 (2011), 180–185. doi:10.1002/wics.147
- [53] Kanit Wongsuphasawat, Zening Qu, Dominik Moritz, Riley Chang, Felix Ouk, Anushka Anand, Jock Mackinlay, Bill Howe, and Jeffrey Heer. 2017. Voyager 2: Augmenting Visual Analysis with Partial View Specifications. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. ACM, Denver Colorado USA, 2648–2659. doi:10.1145/3025453.3025768
- [54] Yilin Ye, Jianing Hao, Yihan Hou, Zhan Wang, Shishi Xiao, Yuyu Luo, and Wei Zeng. 2024. Generative AI for visualization: State of the art and future directions. *Visual Informatics* 8, 2 (June 2024), 43–66. doi:10.1016/j.visinf.2024.04.003
- [55] Ji Soo Yi, Youn ah Kang, John Stasko, and J.A. Jacko. 2007. Toward a Deeper Understanding of the Role of Interaction in Information Visualization. *IEEE Transactions on Visualization and Computer Graphics* 13, 6 (Nov. 2007), 1224–1231. doi:10.1109/TVCG.2007.70515